卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-WF-0508 30-B003701S-S0508

图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

消防 专业 第 05 卷 第 08 册
Fire Fighting SPECIALITY VOLUME BOOK

卷册名称 检修备品库室内消火栓及灭火器布置图
Volume name Maintenance Workshop & Spare Parts Storage Indoor Hydrant & Fire Extinguisher Plane Arrangement Drawing

批 准 APPROVED BY	
审 核 REVIEWED BY	
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图 纸 3 张 DRAWING NOS.	打印件 1 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文 件 名 称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
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		Maintenance Workshop & Spare Parts Storage Indoor Fire Hydrant & Fire Extinguisher				
		Construction & Installation Instructions				
2	30-B003701S-S0508-002	检修备品库 一层室内消火栓及灭火器布置图	A	1		
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3	30-B003701S-S0508-003	检修备品库 室内消火栓及灭火器材料表	A	1		
		Maintenance Workshop & Spare Parts Storage Bill of Materials for Indoor Hydrant & Fire Extinguisher				

注:

1. 本册图采用的主要设计验收规范

1. Design & Acceptance Standards and Codes:

1.1《建筑设计防火规范》GB50016-2014

1.1《Code for fire protection design of buildings》GB50016-2014

1.2《消防给水及消火栓系统技术规范》 GB50974-2014

1.2《Technical code for fire protection water supply and hydrant systems》GB50974-2014

1.3《火力发电厂与变电站设计防火规范》GB50229-2006

1.3《Code for design of fire protection for fossil fuel power plants and substations》GB50229-2006

1.4《建筑灭火器配置设计规范》GB50140-2005

1.4《Code for design of extinguisher distribution in buildings》GB50140-2005

1.5《建筑机电工程抗震设计规范》GB50981-2014

1.5《Code for seismic design of mechanical and electrical equipment》 GB 50981-2014;

1.6 国家建筑标准设计《给水排水标准图集》

1.6 National building standard design "standard collective drawings of water supply and drainage(SCDWSD)"

2.主要设计概况

2. Design Overview

2.1 本册图的设计范围为建设区检修备品库室内消火栓及其管道和灭火器布置图。消防水源自站内室外消防环管上引接，站区采用独立的消防供水系统，泵房内设一台主电动消防泵，一台备用柴油机消防泵和一套气压稳压设备。每台消防泵参数暂定为(尚未招标)：$Q=800m^3/h$,H=90m,室外消防水管网布置图详见本工程S0503卷册。具体接入位置可根据现场情况略做调整，管网工作压力约为0.97MPa。

2.1 The design scope of this volume is the arrangement plan of indoor fire hydrants and mobile fire extinguishers for the Maintenance Workshop & Spare Parts Storage. Fire-fighting water sources from the outdoor fire ring in the station. Independent fire-fighting water supply system is adopted here. There are one(1) 100% capacity motor driven fire-fighting pump, one(1) 100% capacity diesel fire-fighting pump and a set of fire-fighting water pressure maintaining in the fire-fighting water pump house. The parameter)of the fire-fighting pumps is tentative: $Q=800m^3/h$,H=90m. Outdoor fire-fighting networks arrangement plan refers to the volume of S0503 of this project. The specific access location can be adjusted slightly according to the situation, and the pressure of the pipe network is about 0.97MPa.

2.2 站前区检修备品库建筑面积 1426.13m^2，建筑体积约22604.1m^3,丁类仓库，耐火等级为二级。室外消防用水量为25L/s，室内消火栓用水量为10L/s。

2.2 The area of the Maintenance Workshop & Spare Parts Storage is 1426.13m^2. The volume of the building is about 22604.1m^3. ClassⅣ warehouse. Fire resistance rating is class 2. The outdoor fire-fighting water consumption for this building is 25L/s at one time. Water consumption of indoor fire hydrant is 10L/s at one time.

2.3 本册图标高单位以米计，管径及尺寸单位以毫米计，标高采用相对高程系统。图中室内±0.00米标高相当于EGM2008高程系统标高193.85m。室内外高差为0.45m。

2.3 The unit of elevation is m and that of all dimensions (except otherwise specified) is mm. Relative elevationsystem is applied; 0.00m elevation equals 193.85m EGM2008 elevation. Elevation difference between building ground elevation and plant ground elevation is 0.45m. Architecture AB coordinate is used. And for detail information of coordinate system, please refer to plant general layout drawing.

2.5 本图册消防管道标高指管中心标高；室内消火栓标高为栓出口标高，距室内地坪1.10m。

2.5 The elevation of indoor fire-fighting piping in this drawing is pipe central line elevation.Indoor fire hydrant outlet elevation is higher 1.10m than floor ground elevation.

3. 室内管道及阀门安装

3. Indoor pipeline and valve installation:

3.1 系统管网工作压力为0.97MPa，试验压力为1.45MPa。具体的水压试验要求详见相关施工验收规范。

3.1 The fire fighting water piping working pressure is 0.97MPa and its test pressure is 1.45MPa.

3.2 本册图中室内消防管道均采用内外热浸锌镀锌钢管。DN50及以下管径的管道及阀门采用螺纹连接；DN50管径以上的管道及阀门采用刚性沟槽(卡箍)或法兰连接。如采用法兰连接，必须二次镀锌。

3.2 The material of fire fighting piping is hot dip galvanized steel pipe. The fire fighting piping(the diameter is less than 50mm)adopts threaded connection.The fire fighting piping(the diameter is larger than 50mm)adopts clamp connection.

3.3 本册图中所有消防钢管安装时均须进行防腐并涂公安红面漆处理。具体防腐涂料要求见下表或参考涂料厂家资料。安装时应注意防止损坏镀锌和防腐层。

3.3 All steel pipes installed in this volume shall be protected against corrosion and painted with red paint. Specific anti-corrosive coating requirements see below table or reference paint manufacturer data. It is necessary to prevent the damage of galvanizing and anti-corrosion coating when installation.

部位 Position	品种 Type	油漆名称 Paint name	道数 Quan.	漆膜厚度μm Thickness(μm)
埋地钢管 (镀锌钢材) 外壁 Buried steel pipe (galvanized steel) The outer wall	封闭层 Sealing layer	H53-42环氧封闭涂料 H53-42 epoxy resin coating	1	30
	中间层漆 Intermediate coat	842环氧云铁防锈漆 842 epoxy iron antirust paint	1	60
	面漆 Finishing coat	546超厚浆型环氧沥青防锈漆 546 super thick epoxy asphalt antirust paint	1	125
室内架空钢管 (镀锌钢材) 外壁 Indoor overhead steel pipe (galvanized steel) The outer wall	封闭层 Sealing layer	H53-42环氧封闭涂料 H53-42 epoxy resin coating	1	30
	中间层漆 Intermediate coat	842环氧云铁防锈漆 842 epoxy iron antirust paint	1	100
	面漆 Finishing coat	J43-32公安红色氯化橡胶面漆 J43-32 police red chlorinated rubber topcoat	1	50

3.4 管道穿楼板、墙壁应安装套管,套管高出楼层地坪5~10cm，管道与套管间的空隙用柔性不燃材料(玻璃纤维、硅酸铝纤维、岩棉等)封堵。具体做法可参照给水排水标准图集S24。管道在安装过程中如碰到梁、柱及设备管道等情况,可根据现场实际情况弯绕。

3.4 Wall-through bushing must be used when the pipe through the floor or wall, and the top of bushing is 50~100mm higher than ground elevation. The interspace between the pipe and bushing is blocked up by employing flexible noncombustible material (fiberglass、aluminosilicate fiber、rock wool and so on).The location of fire fighting piping could be shifted or bended slightly to avoid the collision with beam、column and equipments.

3.5 当管道穿过防火墙时，管道与防火墙之间的缝隙应采用柔性防火材料堵塞。当直径大于等于32mm的管道穿过防火墙时，除填塞防火材料外，还应采取阻火措施。措施包括采用阻火圈或阻火带并辅以如防火泥或防火密封胶的有机堵料等封堵。

3.5 When the pipeline passes through the firewall, the gap between the pipe and the firewall should be blocked by fire retardant materials. When the diameter of the pipeline is greater than or equal to 32mm through the firewall,in addition to filling fire materials, but also should take measures to prevent fire.

3.6. 管道在安装过程中如碰到梁、柱及设备管道等情况,可根据现场实际情况弯绕，注意不影响交通和美观。

3.6 If the fire-fighting pipelines encounter beams, columns and equipment pipelines etc. in the process of installation, they can be adjusted according to the actual situation. Attention not to affect traffic and aesthetics.

3.7. 为排出管网中的空气，消防管道的最高点或脊峰处设自动排气阀，具体位置详见本卷册消防管道系统图，也可根据现场情况确定。

3.7 The automatic exhaust valve shall be installed at the end and the highest point of the water supply pipe network and the pipe section with obvious fluctuations in order to eliminate air. The set position see the Fire piping system diagram of this volume, while it can also be determined according to the site conditions.

3.8 消防进水埋地管不宜穿越建筑基础，当必须穿越时，应采取防护套管等保护措施。埋地管道的敷设要求参见站区室外消防水管。

3.8 The fire-fighting water should not cross the building foundation. The protective casing shall be adopted when crossing. The laying of buried pipelines is required to see the outdoor fire-fighting water networks layout requirements.

3.9 室内检修隔离阀门应选用明杆闸阀或带启闭刻度的暗杆闸阀，材质为球墨铸铁。

3.9 Indoor inspection and isolation valve shall be operated with a bright stem gate valve or a dark stem gate valve with open and closed scale. The material is nodular cast iron.

3.10 阀门的布置应方便操作并不影响交通，现场可根据情况略做调整，但请注意不要破环系统隔离原则。

3.10 The arrangement of the valve should be convenient to operate without affecting the traffic. The position can be adjusted slightly according to site conditions, but the principle of system isolation shall not be broken.

3.11 消防给水及消火栓系统在管网安装完毕后，应对其进行强度试验、冲洗和严密性试验。具体试验冲洗要求详见《消防给水及消火栓系统技术规范》GB50974-2014第12.4节。

3.11 After the pipe network is installed, the strength test, flushing and tightness test shall be performed on the system. Specific test flushing requirements refer to《Technical code for fire protection water supply and hydrant systems》GB50974-2014 in section 12.4.

4. 抗震设计

4. Aseismatic design

4.1. 根据本工程岩土资料，拟建场地的地基液化等级均判为中等液化，因此设计时考虑地基土的地震液化影响。

4.1. According to the geotechnical data of this project,that the proposed site were jailed for medium liquefied foundation liquefaction grade, the effects of earthquake liquefaction of foundation soil are considered in design.

4.2. 本工程抗震设防烈度为7度，机电设备及管道的抗震措施应符合规范《建筑机电工程抗震设计规范》GB50981-2014。

4.2. The seismic fortification intensity of this project is 7 degrees, and the anti-seismic measures of mechanical and electrical equipment and pipelines should conform to the specification for seismic design of building mechanical and electrical engineering (GB50981-2014).

4.3. 管道穿过内墙或楼板时，应设置套管;套管与管道间的缝隙，应采用柔性防火材料封堵。

4.3. When the pipe runs through the wall or floor, the casing shall be set; the gap between casing and pipe should be sealed with flexible fireproof material.

4.4. 管道支架需符合《建筑机电工程抗震设计规范》GB50981-2014中的相关规定。

4.4. The pipe supports shall conform to the relevant provisions in gb50981-2014 of the design specification for seismic design of mechanical and electrical engineering.

4.5. 运行时不产生振动的给水水箱、太阳能设备等设备、设施应与主体结构牢固连接，与其连接的管道应采用金属管道。

4.5. The equipment and facilities such as water tanks and solar equipments that do not produce vibration during operation shall be connected with the main structure of the main structure, and the pipe connected to it shall be in metal pipe.

5. 支吊架

5. Pipe support:

5.1 单管立式支架图可参见《给排水标准图集》:03S402页77、78、79、80;

5.1 Single pipe vertical support drawing refer to SCDWSD:03S402 page(77、78、79、80);

5.2 单管托架图可参见《给排水标准图集》:03S402页60、61;

5.2 Single pipe bracket drawing refer to SCDWSD:03S402 page(60、61);

5.3 管路的固定应根据不同情况,分别采用支架或吊架固定,干管尽量用支架固定在建筑结构上(如墙或柱),配水管和支管可采用吊架进行固定。架空管道每段管道设置的防晃支架不应少于1个；当管道改变方向时，应增设防晃支架；立管应在其始端和终端设防晃支架或采用管卡固定。

5.3 Piping shall be fixed using support or hanger according to the different situation. Main pipe shall be fixed to building structure(wall or column)by using support, and branch pipe could use hanger. No less than 1 anti sway bracket shall be set on overhead pipeline. Anti sway bracket should be added when the pipeline changes direction. The riser shall be fixed by setting anti-sway braket or pipe clamp at both of the beginning and end of the riser.

6. 室内消火栓安装

6. Indoor Fire hydrant installation:

6.1 墙体上安装的消火栓箱采用膨胀螺栓固定,设置在钢柱或者设在钢于台上的消火栓箱采用支架焊接。本楼消火栓均半暗装；消火栓及箱体安装等详见建筑专业图纸以及《给水排水标准图集》15S202页19、57、58。

6.1 The fire hydrant box is installed in wall adopts the expansion bolt. Fire hydrant box could be fixed in a steel column or in steel platform with brackets jointing. The fire hydrants in this build are all semi-conceal intalled. Fire hose reel and fire hydrant box drawing refer to architectural drawings and SCDWSD:15S202,page 19/57/58.

6.2 消火栓采用单出口消火栓,栓口及水龙带直径为DN65，水龙带长25m，配D19直流/喷雾两用水枪；消防软管卷盘应配置内径不小于φ19的消防软管,长度为30m。

6.2 Fire hydrant with single outlet is set D19 DC/spray water gun. The diameter of hydrant outlet is DN65, water hose length is 25m.Fire hose reel should be equipped with fire hose diameter is not less than 19mm of the length of up to 30m.

6.3 每个消火栓处设消防报警按钮，报警信号接至火灾探测报警系统。

6.3 Each fire hydrant is equipped with fire alarm button, alarm signals are connected to the fire detection and alarm system.

6.4 本册图的室内消火栓均应采用SNW65-Ⅲ型减压稳压消火栓,减后栓口动压应控制在0.35~0.5MPa之间。见给水排水标准图集15S202页62。

6.4 The indoor fire hydrant in this volume shall use the SNW65-Ⅲ type pressure reducing and stabilizing hydrant, and the dynamic pressure of the hydrant should be controlled between 0.35~0.5MPa. Fire hydrant drawing refer to SCDWSD:15S202,page(62).

6.5 由消火栓XF1兼作试验消火栓，带压力表。见给水排水标准图集15S202页54。试验时可用龙带将水排至室外屋面排水沟或卫生间下水道。

6.5 The fire hydrant XF1 is used as the test fire hydrant, with pressure gauge. See SCDWSD:15S202,page(54). The test can be used to bring water to the outdoor roof drainage ditch or toilet drain.

7. 移动式灭火器

7. Fire extinguisher:

7.1 建筑物内应根据《建筑灭火器配置设计规范》GB50140-2005要求配置移动式灭火器。材料库检修楼建筑火灾以A类为主，火灾危险等级均为中级。

7.1 The building should follow the requirements of <Code for design of extinguisher distribution in buildings>(GB50140-2005), the firetype is class A,the dangerous grade are middle class.

7.2 灭火器配置详见图示，放置时可略微调整。但移动式灭火器应设于位置明显、便于取用且不影响安全疏散的位置。其它有关灭火器的摆放要求见GB50140-2005第5.1条。

7.2 Mobile fire extinguisher should set at the obvious place、convenient to use and no effection to safe evacuation, refer to GB50140-2005 5.1.

7.3 本册图中安装在消防柜中的灭火器不再表示(每个消防柜放置4kg手提式干粉磷酸铵盐灭火器2具)，仅表示额外需另增加的部分。额外增加的每2具手提式灭火器配1只移动式灭火器箱。

7.3 The fire extinguishers installed in the fire box are no longer indicated (each fire cabinet places 2 sets of 4kg portable dry powder ammonium phosphate fire extinguisher). Only the additional part is expressed. 2 additional portable fire extinguishers are put in 1 mobile fire extinguisher cabinet.

8. 图例(Graphic symbols):

——	消防水管道	Firewater pipe
⋈	闸阀	Gate valve
▷	大小头	Reducer
◪	室内消火栓	Indoor fire hydrant
XHL-1	消防立管编号	Fire riser serial number
XF	消火栓编号	Fire hydrant serial number
ZF	消火栓管道隔离阀门编号	Hydrant pipe isolation valve serial number
▲	磷酸铵盐干粉灭火器	Ammonium phosphate powder extinguisher

数量
Number

2×MF/ABC4 — 型号 Type
2(A)
单具灭火级别
Single extinguishing level

Equips ammonium phosphate powder extinguishers(MF/ABC4),two in each box.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

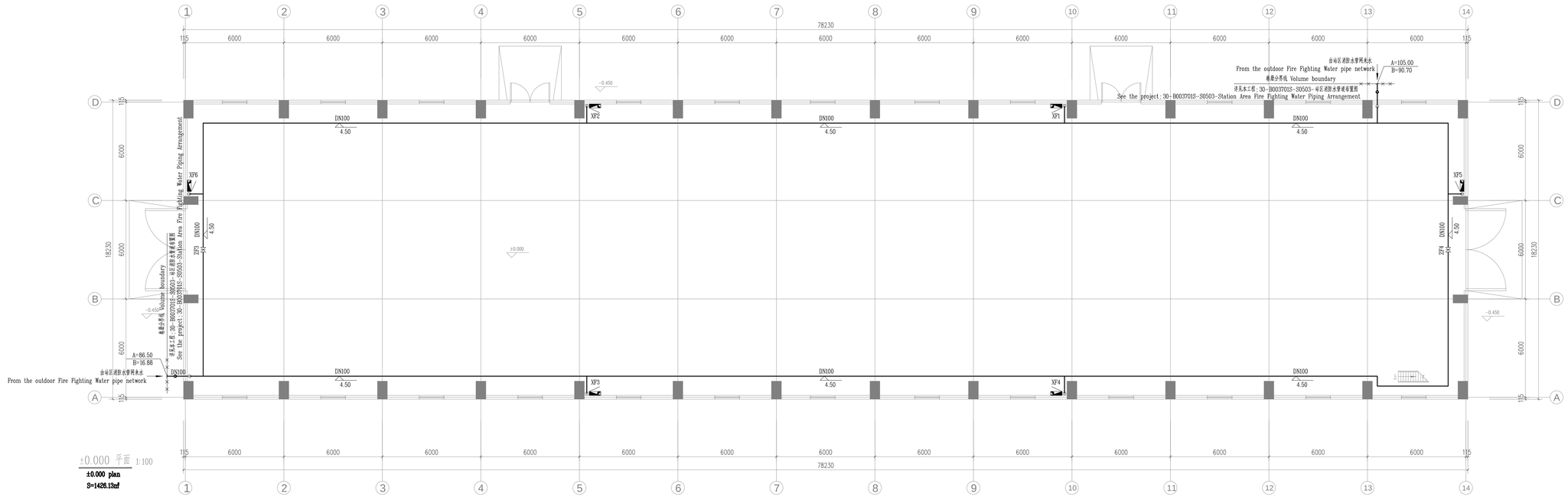
M L H V D C - L C S - E C I - D D - W F - 0 5 0 8 - 0 0 1 A

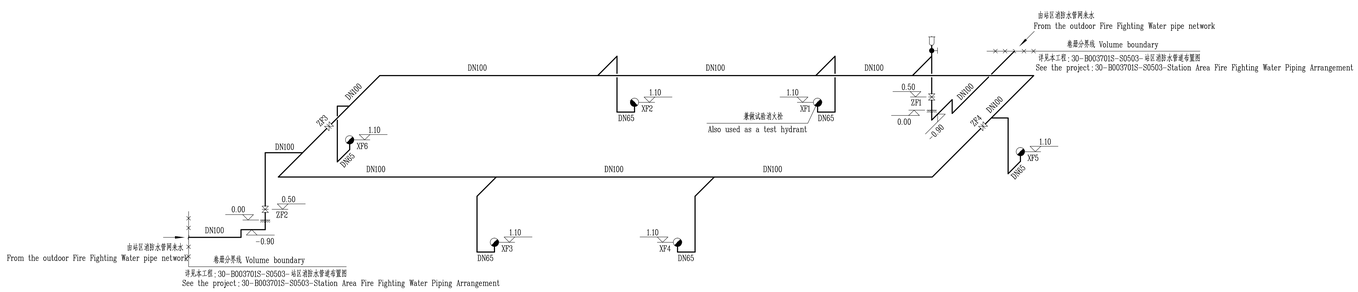
	姓名 Name	日期 Date
设计 Designed	宋晨晓	2018.01.11
校核 Checked	李科杰	2018.01.11
审核 Reviewed	陈计甲 陈计甲	2018.01.11
批准 Approved	陈计甲 陈计甲	2018.01.11

检修备品库
Maintenance Workshop & Spare Parts Storage
室内消火栓及灭火器施工安装说明
Indoor Fire Hydrant & Fire Extinguisher Construction & Installation Instructions

比例 Scale: 1:100
图号 DWG NO.: 30-B003701S-S0508-001
版本 REV.: A

594X841





室内消火栓透视图
Indoor Hydrant Perspective Drawing

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦联网至亚里—拉合尔±660kV直流输电工程
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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - W F - 0 5 0 8 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed	宋玉成	2018.01.11
校核 Checked	谭耕杰	2018.01.11
审核 Reviewed	邵世平 陈俊	2018.01.11
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检修备品库
Maintenance Workshop & Spare Parts Storage
一层室内消火栓、灭火器布置图及透视图
The First Floor Indoor Hydrant & Fire Extinguisher Plane Arrangement Drawing & Perspective Drawing

比例 Scale: 1:100 | 图号 DWG. NO.: 30-B003701S-S0503-002 | 版本 REV.: A

841X1189

编号 No.	名 称 Name	规 格 Parameters	单位 Unit	数量 Quan	材料 Material	单重 Single 重 量 (公斤)	总重 Totoal Weight(Kg)	参考图号及备注 Remark
18	截止阀	DN25 , PN=1.6MPa	只	1				
	global valve		Set					
17	排气阀	DN25 , PN=1.6MPa	只	1				
	vent valve		Set					
16	防腐油漆	实报实销						
	Anti-corrosive coating	Accountable	Set					
15	管道吊支架	每套按8公斤计	套	50				
	hanger and bracket	8kg each	Set					
14	法兰	DN100 PN=1.6MPa	只	8	球墨铸铁			
	flange		Set		Nodular cast iron			
13	卡箍接头	DN65 PN=1.6MPa	只	30	球墨铸铁			
	Clamp joint		Set		Nodular cast iron			
12	卡箍接头	DN100 PN=1.6MPa	只	44	球墨铸铁			
	Clamp joint		Set		Nodular cast iron			
11	卡箍三通	DN100/DN65 PN=1.6MPa	只	6	球墨铸铁			
	Clamp Tee		Set		Nodular cast iron			
10	卡箍三通	DN100/DN100 PN=1.6MPa	只	2	球墨铸铁			
	Clamp Tee		Set		Nodular cast iron			
9	90°卡箍弯头	DN65 PN=1.6MPa	只	12	球墨铸铁			
	90°clamp elbow		Set		Nodular cast iron			
8	90°卡箍弯头	DN100 PN=1.6MPa	只	8	球墨铸铁			
	90°clamp elbow		Set					
7	手动闸阀	DN100 PN=1.6MPa	只	4				Z45T-16型暗杆楔式(带启闭刻度)
	Manual Gate valve		Set					Z45T-16,non-rising stem wedge(including startup scale)
6	内外热浸镀锌钢管	D33.7×3.2(DN25),PN=1.6MPa	米	3	Q235B			GB/T 3091-2015
	Hot dip galvanized steel pipe		m					
5	内外热浸镀锌钢管	D114.3×4(DN100),PN=1.6MPa	米	200	Q235B			GB/T 3091-2015
	Hot dip galvanized steel pipe		m					
4	内外热浸镀锌钢管	D76.1×4(DN65),PN=1.6MPa	米	20	Q235B			GB/T 3091-2015
	Hot dip galvanized steel pipe							
3	灭火器	MF/ABC4磷酸铵盐灭火器	个	12				
	Fire extinguisher	MF/ABC4 Ammonium phosphate fire extinguisher	Set					
2	减压稳压消火栓(含消火栓箱)	SNW65-III型减压稳压消火栓, PN=1.6MPa	套	5				给水排水标准图集15S202/19、57、58
	SNW65-III hydrant Including fire hydrant box	SNW65-III pressure relief and stablizing hydrant PN=1.6MPa	Set					ARCHLIB 15S202/19、57、58
1	试验消火栓 (含消火栓箱)	SNW65-III型减压稳压消火栓(带压力表) PN=1.6MPa	套	1				给水排水标准图集15S202/19、57、58
	Test hydrant Including fire hydrant box	SNW65-III pressure relief and stabilize hydrant PN=1.6MPa, including pressure gage	Set					ARCHLIB 15S202/19、57、58

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - W F - 0 5 0 8 - 0 0 3 A

	姓名 Name	日期 Date
设 计 Designed	朱晓晓	2018.01.11
校 核 Checked	路科杰	2018.01.11
审 核 Reviewed	陈沂亨	2018.01.11
批 准 Approved	陈沂亨	2018.01.11

检修备品库

Maintenance Workshop & Spare Parts Storage

室内消火栓及灭火器材料表

Bill of Materials for Indoor Hydrant & Fire Extinguisher

比例 Scale: 1:100

图号 DWG NO.: 30-B003701S-S0508-003

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